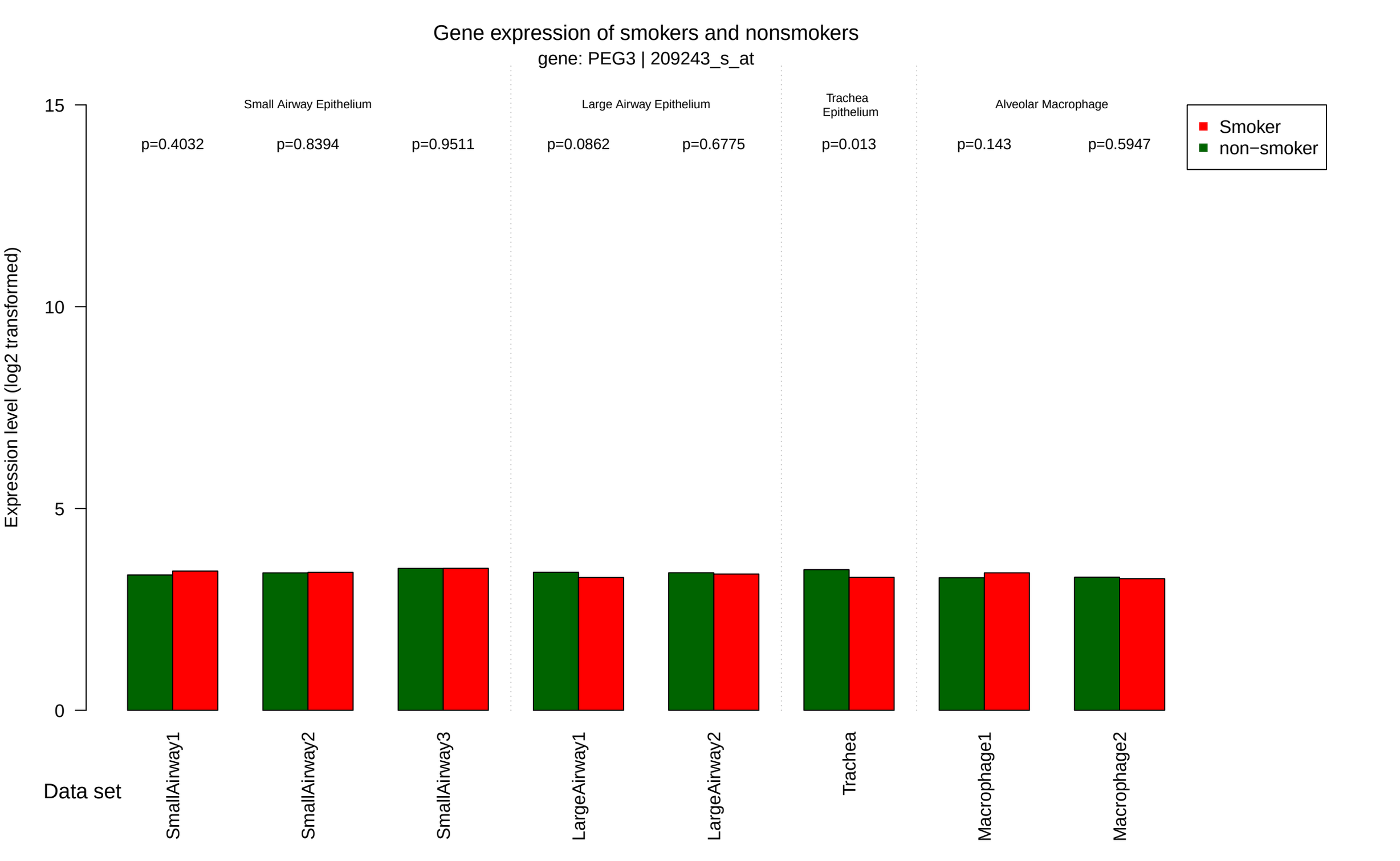
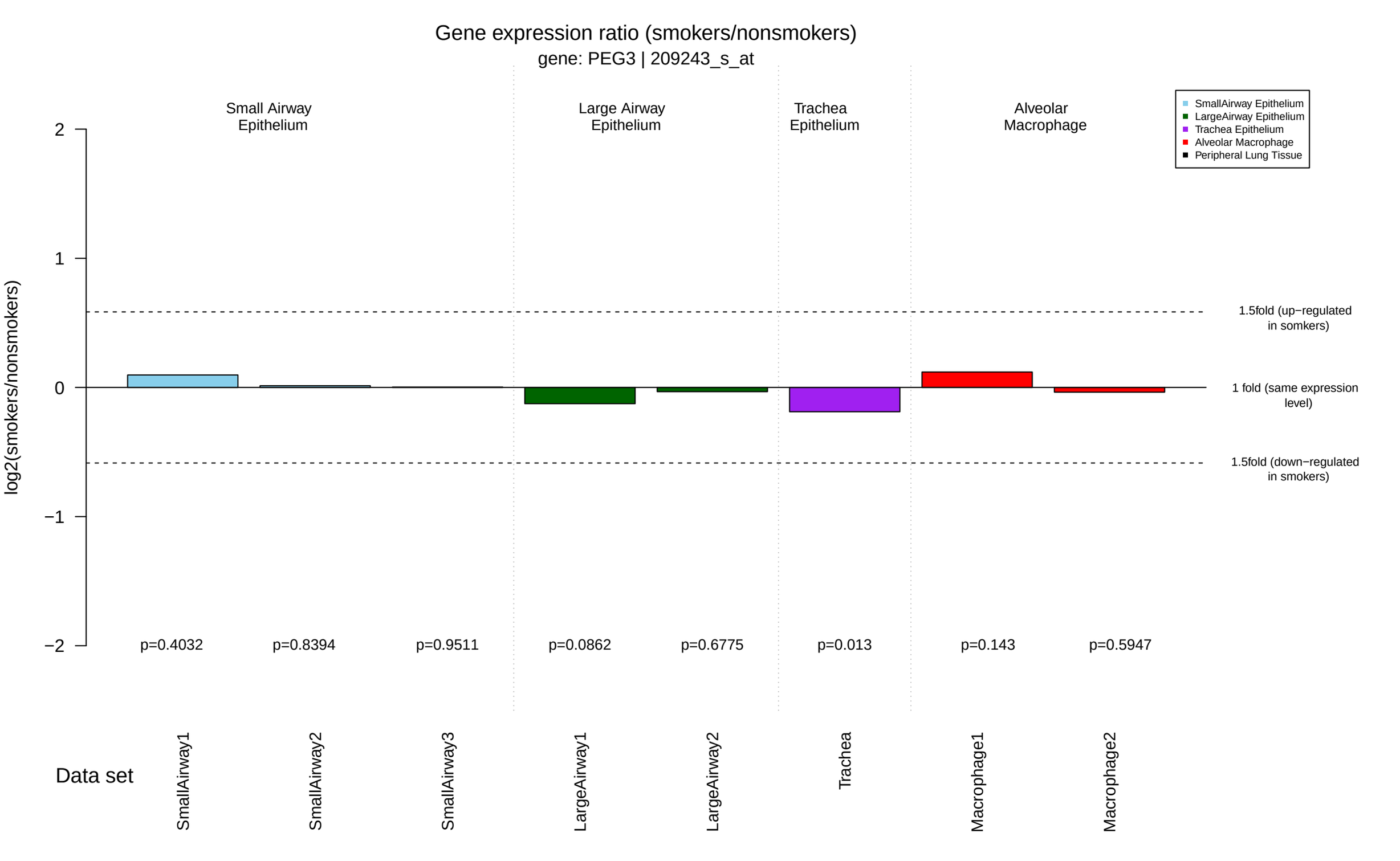
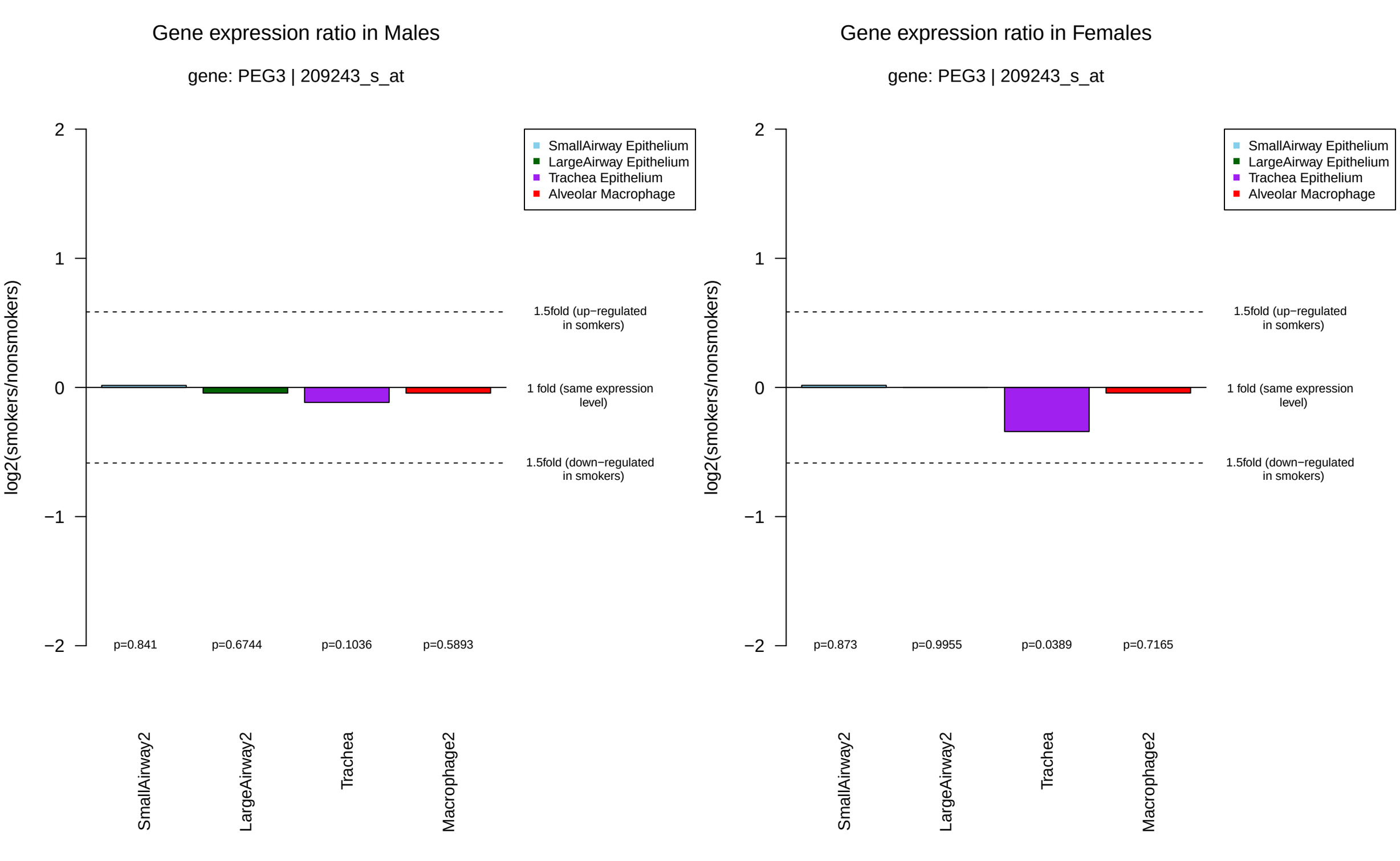
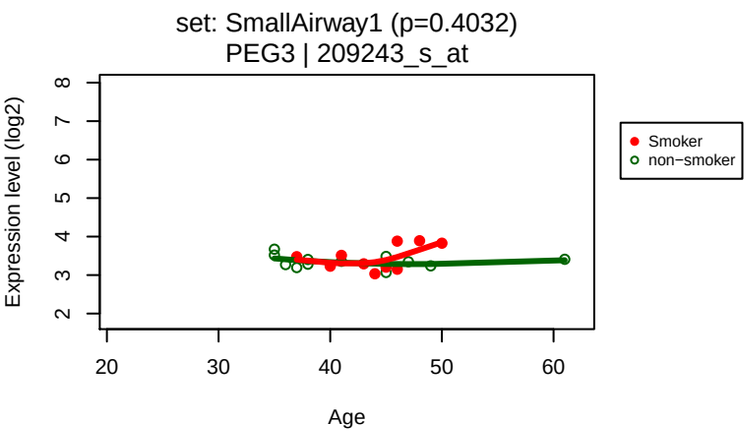
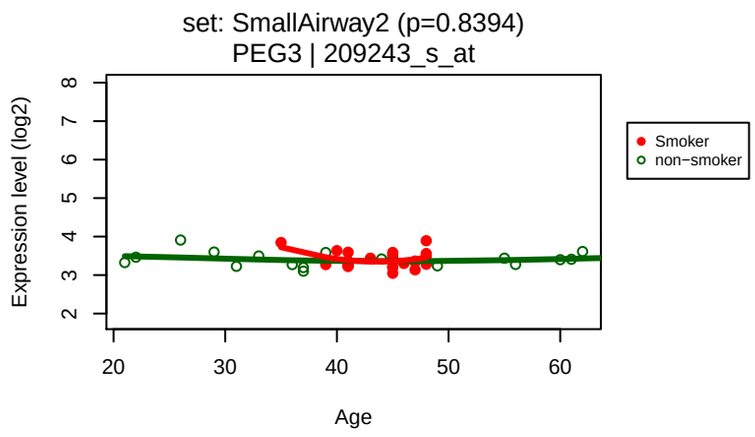
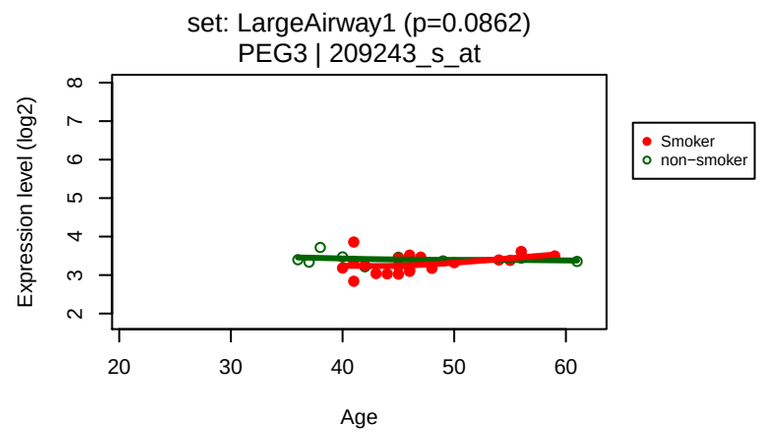

Gene expression of smokers and nonsmokers
gene: PEG3 | 209243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4032
p=0.8394
p=0.9511
p=0.0862
p=0.6775
p=0.013
p=0.143
p=0.5947
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

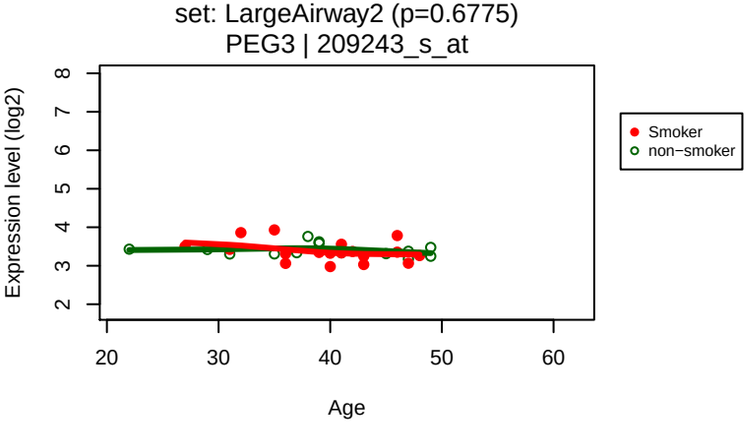

Gene expression ratio (smokers/nonsmokers)
gene: PEG3 | 209243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4032
p=0.8394
p=0.9511
p=0.0862
p=0.6775
p=0.013
p=0.143
p=0.5947
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

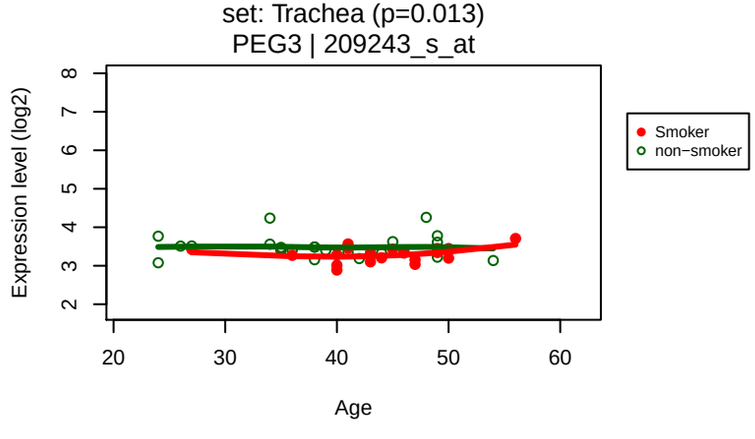

Gene expression ratio in Males
gene: PEG3 | 209243_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.841
p=0.6744
p=0.1036
p=0.5893
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEG3 | 209243_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.873
p=0.9955
p=0.0389
p=0.7165
SmallAirway2
LargeAirway2
Trachea
Macrophage2

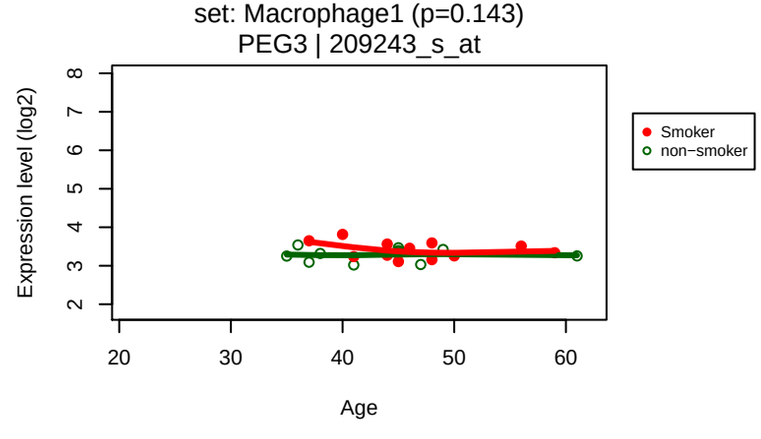

set: SmallAirway1 (p=0.4032)
PEG3 | 209243_s_at
Expression level (log2)
Age
Smoker
non−smoker

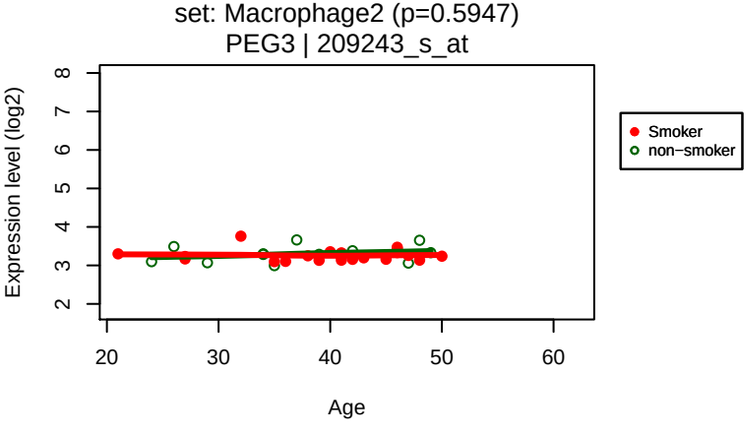

set: SmallAirway2 (p=0.8394)
PEG3 | 209243_s_at
Expression level (log2)
Age
Smoker
non−smoker

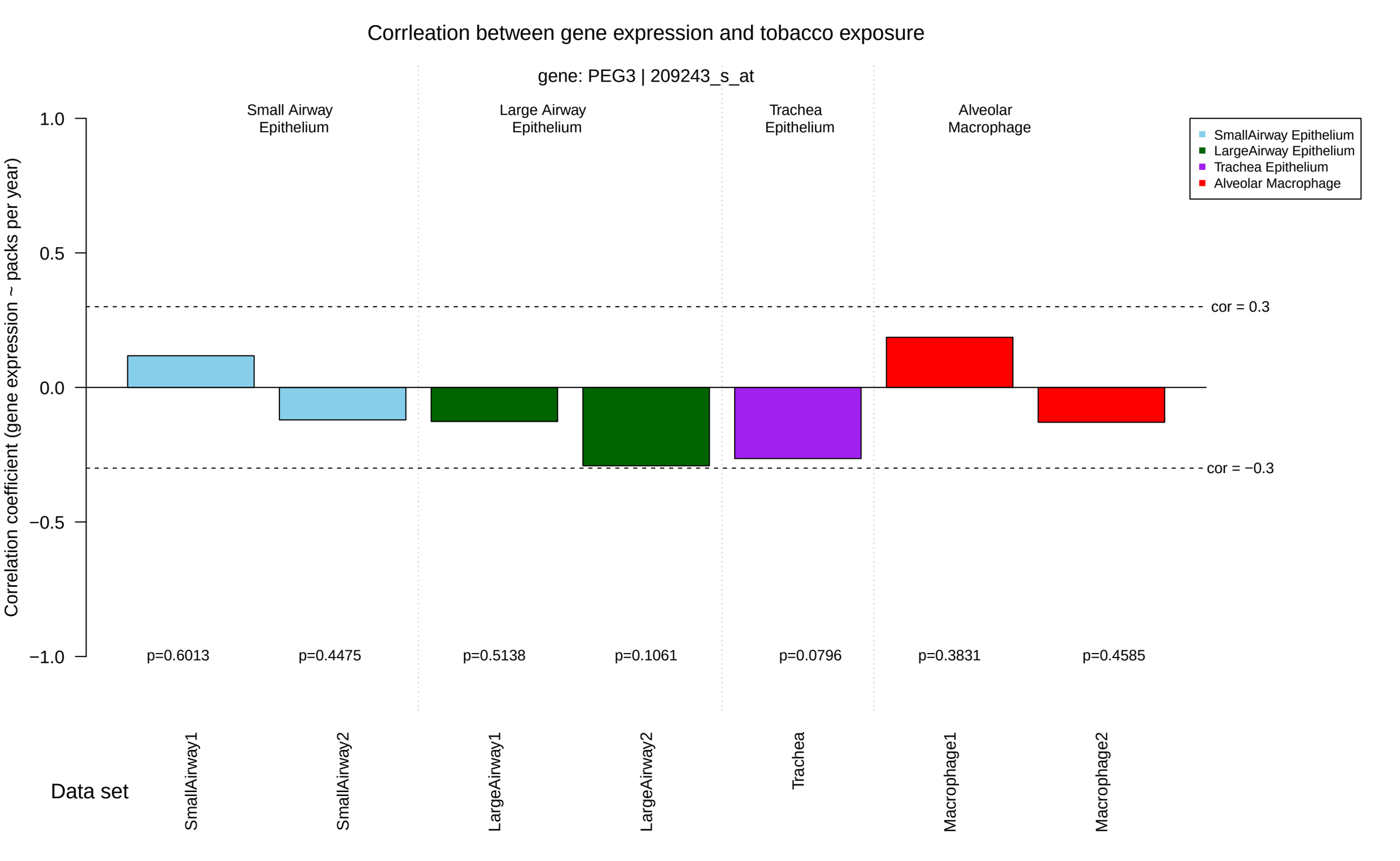

set: LargeAirway1 (p=0.0862)
PEG3 | 209243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6775)
PEG3 | 209243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.013)
PEG3 | 209243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.143)
PEG3 | 209243_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5947)
PEG3 | 209243_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEG3 | 209243_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6013
p=0.4475
p=0.5138
p=0.1061
p=0.0796
p=0.3831
p=0.4585
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2